

Design example traverse crane

$$G(s) = \frac{1.5}{s^2 + 0.4s + 1.5} \quad G_w = \frac{2}{3}$$

- Spec:
- no overshoot for reference steps
 - settling time improved by 20%
 - not affected by a constant wind disturbances.

I. Analysis of the spec. and admissible pole region:

a) no overshoot
 $\Rightarrow$ poles have to be on the real axis

b) current settling time

$$G'(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$1\% - t_s : t_s = \frac{4.6}{\zeta\omega_n} = \frac{4.6}{0.2} = 23 \text{ sec}$$

Reduction bei 20%

$$\Rightarrow t_s = \frac{18.4}{\cancel{18.4}} \text{ sec for the } 1\% \text{ level.}$$

$$\Rightarrow \zeta = \frac{4.6}{18.4} = 0.25$$

$\Rightarrow$ poles have to be left of -0.25 on the real axis

c) Wind disturbance requirement

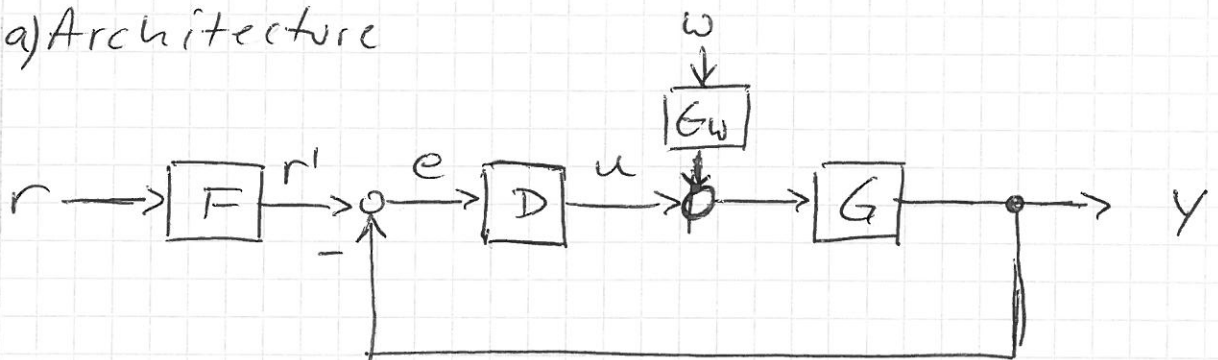
$$\Rightarrow e_{ss} = 0 \text{ for disturbances.}$$

$\Rightarrow$ a free integrator term is needed in the loop transfer function

To conclude: - pole region $(-\infty, -0.25)$
- integral action needed.

II. Design of the closed loop

a) Architecture



There are two important transfer functions from the specification

$T: r \rightarrow y$, $T_w: w \rightarrow y$

$$\begin{aligned}
 y &= G \cdot u + G \cdot G_w w \\
 &= G \cdot D (F r - y) + G \cdot G_w w \\
 &= G \cdot D \cdot F r - G \cdot D y + G \cdot G_w w
 \end{aligned}$$

$$\Rightarrow y = \underbrace{\frac{G D F}{1 + G D}}_T \cdot r + \underbrace{\frac{G \cdot G_w}{1 + G D}}_{T_w} w$$

tracking
response

disturbance response

~~b) choice of controller D~~
~~We select a PID $\Rightarrow D(s) = \frac{k_D s^2 + k_P s + k_I}{s}$~~

there are two more important transfer functions: $r \rightarrow u$, $w \rightarrow u$

Why? - The actuator might have limitations.

$$\begin{aligned} u &= D \cdot (F \cdot r - y) \\ &= D \cdot F \cdot r - D(G \cdot u + G_G w) \\ &= D \cdot F \cdot r - D G \cdot u - D G \cdot G_G \cdot w \end{aligned}$$

$$\begin{aligned} \Rightarrow u &= \underbrace{\frac{D \cdot F}{1 + D \cdot G}}_{} r - \underbrace{\frac{D G G_G}{1 + D \cdot G}}_{} w \\ &= D \cdot (1 - T) \quad \quad \quad \cancel{D} \cdot T_w \end{aligned}$$

b) Closed loop dynamics with D
Select $D(s)$ as a PID controller

$$D = \frac{k_D s^2 + k_P s + k_I}{s}$$

From slide 14, lecture 9 we know

$$T(s) = \frac{1.5 k_D s^2 + 1.5 k_P s + 1.5 k_I}{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I}$$

$$T_w(s) = \frac{s}{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I}$$

Now we can also derive $D(1-T)$ and $D T_w$

$$\Phi \cdot (1-T)$$

$$= \frac{k_D s^2 + k_P s + k_I}{s}$$

$$\frac{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I - 1.5 k_D s^2 - 1.5 k_P s - 1.5 k_I}{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I}$$

$$= \frac{(k_D s^2 + k_P s + k_I) (s^3 + 0.4 s^2 + 1.5 s)}{s (s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I)}$$

$$= \frac{k_D s^4 + (k_P + 0.4 k_D) s^3 + (1.5 k_D + 0.4 k_P + k_I) s^2 + \dots}{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I}$$

$$\dots (1.5 k_P + 0.4 k_I) s + 1.5 k_I$$

$$\Phi \cdot T_W$$

$$= \frac{k_D s^2 + k_P s + k_I}{s^3 + (0.4 + 1.5 k_D) s^2 + (1.5 + 1.5 k_P) s + 1.5 k_I}$$

Note:

- same denominator as before
- very different numerators
- $\Phi(1-T)$ has a numerator of higher order than the denominator.

Analysis of these transfer functions give a full understanding of the loop.

III. Determining an appropriate controller and loop behaviour.

a) Choose $p_{1,2,3} = -0.25$

$$\begin{aligned}\hat{P}(s) &= (s + 0.25)^3 \\ &= s^3 + 0.75s^2 + 0.1875s + 0.015625\end{aligned}$$

$$P(s) = s^3 + (0.4 + 1.5k_D)s^2 + (1.5 + 1.5k_P)s + 1.5k_I$$

compare coefficients

$$\begin{aligned}s^2: 0.4 + 1.5k_D &= 0.75 \\ \Rightarrow k_D &= \frac{0.75 - 0.4}{1.5} = \frac{0.35}{1.5} \approx 0.233\end{aligned}$$

$$\begin{aligned}s^1: 1.5 + 1.5k_P &= 0.1875 \\ \Rightarrow k_P &= \frac{0.1875 - 1.5}{1.5} = -0.875\end{aligned}$$

$$\begin{aligned}s^0: 1.5k_I &= 0.015625 \\ \Rightarrow k_I &= 0.01041\end{aligned}$$

Warning: k_P is negative!

Let's review the transfer functions, and simulate the closed loop behaviour.

What went wrong?

- RHP zero in T and zero on the imaginary axis for T_w
- Remember the effect of zeros.

b) Choose $p_{1,2,3} = -1$

As we could see from the coefficient comparison, k_P was negative. This can be avoided by a better choice of the poles.

$$\begin{aligned}\hat{P}(s) &= (s+1)^3 \\ &= (s^2+2s+1)(s+1) \\ &= s^3 + 3s^2 + 3s + 1\end{aligned}$$

Again we compare coefficients and derive controller parameters

$$\begin{aligned}s^2: \quad 0.4 + 1.5 k_D &= 3 \\ \Rightarrow k_D &= \frac{2.6}{1.5} = 1.733\end{aligned}$$

$$\begin{aligned}s^1: \quad 1.5 + 1.5 k_P &= 3 \\ \Rightarrow k_P &= \frac{1.5}{1.5} = 1\end{aligned}$$

$$\begin{aligned}s^0: \quad 1.5 k_I &= 1 \\ \Rightarrow k_I &= 0.667\end{aligned}$$

Again, • review the transfer functions
• review the closed loop behavior

Much better response. Still, there is an overshoot like behavior, due to the zeros. Can we do anything about this?

Moreover, look at the initial behavior of the carrier! - Is this good?

c) Pre-filter design F

At the moment $F=1$ in

$$T = \frac{GD}{1+GD} \cdot F$$

$$= \frac{1.5k_D s^2 + 1.5k_P s + 1.5k_I}{s^3 + (0.4 + 1.5k_D)s^2 + (1.5 + 1.5k_P)s + 1.5k_I} \cdot F$$

What if we choose

$$F = \frac{1.5k_I}{1.5k_D s^2 + 1.5k_P s + 1.5k_I}$$

$$\Rightarrow T = \frac{1.5k_I}{s^3 + (0.4 + 1.5k_D)s^2 + (1.5 + 1.5k_P)s + 1.5k_I}$$

But: $1.5k_D s^2 + 1.5k_P s + 1.5k_I$
must have all roots in the
left half plane.

In this way we have compensated for
the zeros which create:

- overshoot behavior
- problematic actuator signals.

Note: We cannot affect the disturbance
response using a pre-filter.

d) Relaxing the pre-filter design

Now the loop is responding faster, than necessary.

We could make the pre-filter "slower", such that ~~it~~ fulfills our settling time requirement.

the closed loop

This has to be done iteratively!

$$F = \frac{1}{(2s+1)^2}$$

Result: • Tracking performance fulfilled

- no overshoot for setpoint changes
- moderate control signals